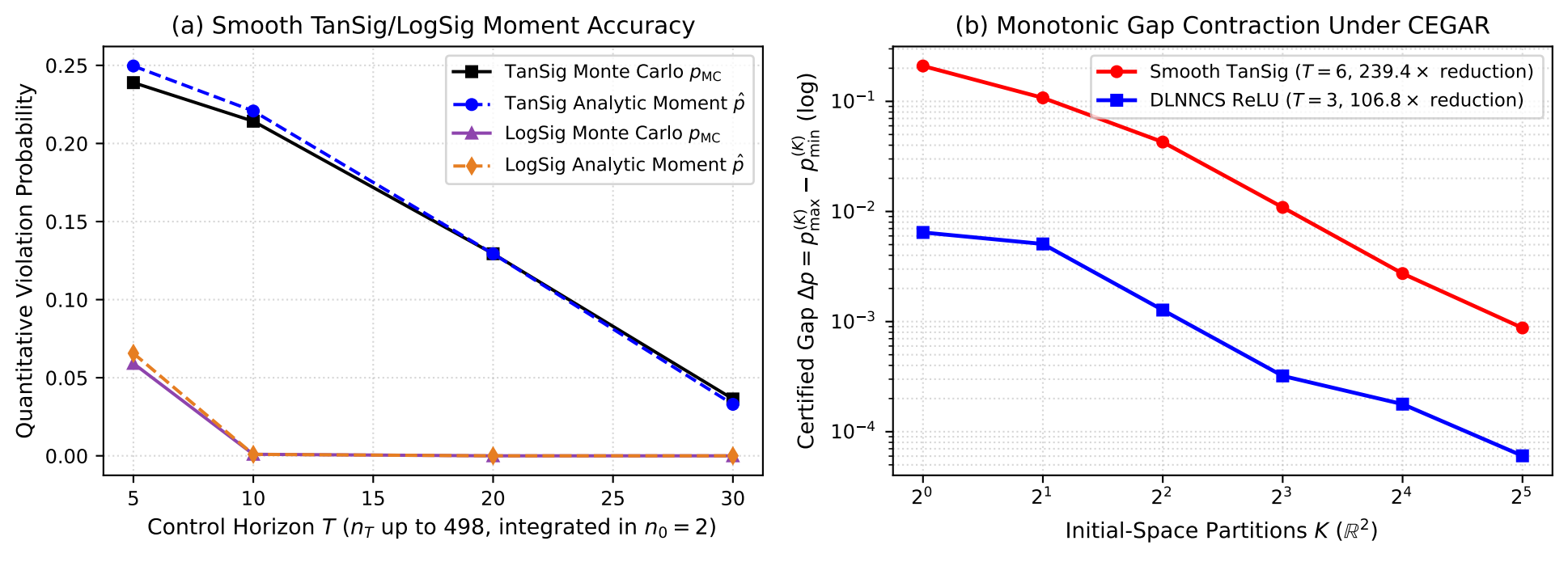
(a) Smooth TanSig/LogSig Moment Accuracy
Quantitative Violation Probability
0.25
0.20
0.15
0.10
0.05
0.00
5
10
15
20
25
30
Control Horizon T (n_T up to 498, integrated in $n_0 = 2$)
TanSig Monte Carlo p_{MC}
TanSig Analytic Moment $\hat{p}$
LogSig Monte Carlo p_{MC}
LogSig Analytic Moment $\hat{p}$
(b) Monotonic Gap Contraction Under CEGAR
Certified Gap $\Delta p = p_{\max}^{(K)} - p_{\min}^{(K)}$ (log)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
2^0
2^1
2^2
2^3
2^4
2^5
Initial-Space Partitions K ($\mathbb{R}^2$)
Smooth TanSig ($T = 6$, 239.4 × reduction)
DLNNCS ReLU ($T = 3$, 106.8 × reduction)